Where Should We Open Our Next Daycare?

MGT-403 Data Project · Pre-Analysis (Prices and Costs) and Final Project Analysis

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Summary

- **Analysis dataset.** 8,146 year × county observations (2015–2022) for 1,159 metropolitan counties with a non-missing NDCP center-based preschool price and every predictor; 998 counties in 2022.
- **Prices and costs move together.** In 2022, a $1,000 higher childcare-worker wage goes with a $10.15/week higher preschool price (R^2 = 0.55); a $100 higher median gross rent goes with $12.05/week ($R^2$ = 0.46). Together: $7.09 and $5.91 ($R^2$ = 0.61). Both shrink in the joint model because wages and rents are correlated (r = 0.67).
- **Recommendation.** Ten counties in ten metros whose prices sit far above what their local wages and rents predict: Stearns (MN), Norfolk (MA), McLean (IL), Howard (MD), King (WA), DuPage (IL), Champaign (IL), Montgomery (MD), Worcester (MA), Washtenaw (MI). Average 2022 premium $85/week.
- **Reliability.** The benchmark model (price predicted from wages and rents) fits held-out metros almost as well as in sample (R^2 0.60 vs 0.61). Premiums are persistent (correlation 0.93 one year apart, 0.55 seven years apart). In a 2018 back-test, the 6 picks that still had 2022 prices were $70/week above their predicted price on average, and all 6 were above it.

Pre-Analysis 1. The analysis dataset

We merged the NDCP county prices (2015–2022) onto the Problem Set 5 year × county metro panel (OEWS childcare-worker wages crosswalked to counties, plus ACS 5-year county data) on the 5-digit FIPS code and year. Every metro county-year in the PS5 panel has a matching NDCP row. The **analysis dataset** keeps county-years where the median center-based preschool price (MCPRESCHOOL) and all ten predictors are non-missing.

Year	Metro counties	Preschool price reported	Wage non-missing	All predictors non-missing	Analysis dataset	Price but a predictor missing
2015	1,233	1,003	1,222	1,221	**994**	9
2016	1,233	1,025	1,219	1,219	**1,019**	6
2017	1,234	1,063	1,224	1,224	**1,057**	6
2018	1,233	1,063	1,228	1,228	**1,062**	1
2019	1,235	1,063	1,229	1,229	**1,061**	2
2020	1,235	1,063	1,215	1,215	**1,050**	13
2021	1,235	918	1,222	1,221	**905**	13
2022	1,235	1,013	1,215	1,207	**998**	15
Total					**8,146**	65

Rows with a price but a missing predictor are mostly suppressed OEWS wage cells (55 county-years) and Connecticut in 2022, where the ACS switched from counties to planning regions so the old county FIPS codes no longer match (7 counties). Some states have no county prices in some years (California, North Carolina, Pennsylvania and Virginia in 2021; Indiana and New Mexico in every year), which is why the yearly count varies.

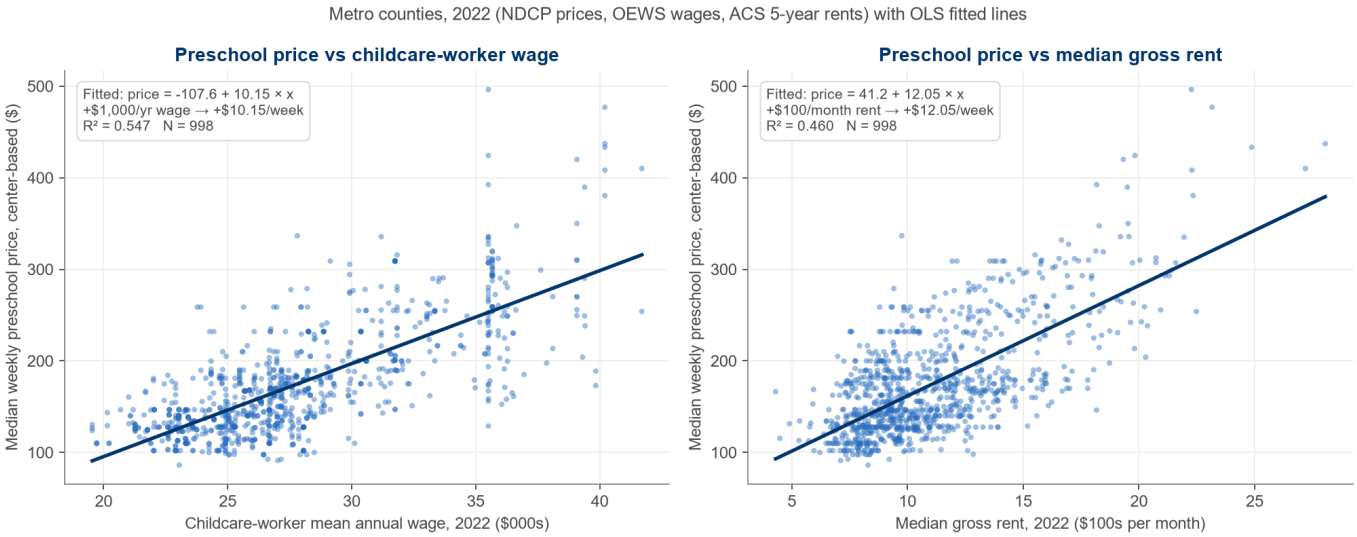
Pre-Analysis 2. Descriptive statistics, 2022

Variable (2022, metro counties)	Mean	SD	N	Min	Median	Max
Median weekly price, center-based preschool ($)	$174.48	$59.75	998	$86.19	$162.12	$496.30
Median weekly price, center-based toddler ($)	$185.47	$65.47	996	$97.00	$170.00	$520.00
Median weekly price, center-based infant ($)	$214.03	$82.62	997	$95.60	$198.93	$606.61
Childcare-worker mean annual wage (OEWS, $)	$27,780	$4,351	998	$19,540	$26,870	$41,700
Childcare-worker mean hourly wage (OEWS, $)	$13.36	$2.09	998	$9.39	$12.92	$20.05
Median gross rent ($/month)	$1,106	$336.40	998	$430.00	$1,019	$2,805
Per-capita income ($)	$37,859	$9,457	998	$16,368	$36,266	$89,702
Total population	255,803	551,110	998	50	99,157	9,936,690
Own children under 6	17,151	37,250	998	0	6,572	616,046
Under-6 w/ all resident parents in labor force	11,538	24,539	998	0	4,286	406,465
Share women 25+ with bachelor's+	30.6%	11.1%	998	8.1%	28.9%	77.6%
Share women 16-64 full-time year-round	44.6%	5.6%	998	21.2%	45.0%	71.4%
Share kids under 6 (of population)	6.4%	1.1%	998	0.0%	6.4%	10.9%
Share kids under 6 who need care (of population)	4.3%	0.9%	998	0.0%	4.4%	7.0%
Childcare-worker employment in the metro area (OEWS)	3,179	6,333	997	30	820	37,900
Counties in the state reporting this exact price (NDCP)	14	22	998	1	4	99

N = 998 counties in 330 OEWS metro areas and 44 states/territories. Prices are full-time weekly medians (NDCP); wages are OEWS mean wages for SOC 39-9011 at the metro-area level, so all counties in one metro share a wage; ACS variables are 2018–2022 5-year estimates. 2022 values are already in 2022 dollars.

Preschool care is the cheapest center-based age group ($174.48/week on average, vs $185.47 for toddlers and $214.03 for infants), and prices are right-skewed (mean above median; maximum $496.30).

Pre-Analysis 3. Prices against wages and rents



Each dot is a metro county in 2022; the line is the OLS fit.

- **Wages.** Prices rise steeply with childcare-worker wages: each $1,000 of mean annual wage goes with $10.15/week more for preschool (correlation 0.74; wages alone explain 55% of price variation). The vertical stacks are counties in the same metro, which share one OEWS wage.
- **Rents.** Prices also rise with rent: each $100/month of median gross rent goes with $12.05/week more (correlation 0.68; $R^2 = 0.46$).
- **Interpretation.** Both relationships fit a market where higher input costs are passed on to families, but neither is causal on its own. Wages and rents are both higher in affluent, dense metros where families can pay more, so each line partly reflects demand. The spread around the lines widens at higher costs (more heteroskedasticity), and some prices form horizontal bands because several states publish one price for a whole rate zone. Counties well above the line are the candidates for the final project.

Pre-Analysis 4. Univariate vs multivariate regressions, 2022

Dependent variable: median weekly center-based preschool price ($). Wage in $1,000s per year; rent in $100s per month. Conventional standard errors in parentheses; standard errors clustered by OEWS metro area (wages are measured at that level) in brackets.

	(a) Wage only	(b) Rent only	(c) Wage + rent
Childcare-worker wage ($000/yr)	**10.153** (0.293) [0.647]		**7.093** (0.367) [0.584]
Median gross rent ($100/mo)		**12.053** (0.413) [0.996]	**5.910** (0.475) [0.922]
Intercept	**-107.587** (8.240) [17.103]	**41.173** (4.779) [9.797]	**-87.946** (7.830) [13.017]
R^2 (adjusted)	0.547 (0.546)	0.460 (0.460)	0.608 (0.607)
Root MSE ($/week)	40.3	43.9	37.5
N	998	998	998

All slope coefficients are significant at $p < 0.001$ with either type of standard error (smallest clustered $t = 6.4$).

Interpretation

- **(a) Wage alone:** counties whose childcare workers earn $1,000 more per year charge $10.15 more per week for preschool, on average.
- **(b) Rent alone:** counties with $100/month higher median gross rent charge $12.05 more per week.
- **(c) Both:** holding rent fixed, $1,000 more in wages goes with $7.09/week; holding wages fixed, $100 more in rent goes with $5.91/week. The two together explain 61% of the variation in 2022 prices, more than either alone.
- **Why the coefficients fall.** Wages and rents are positively correlated (r = 0.67). In a univariate regression each variable also picks up the effect of the omitted one (omitted-variable bias). The decomposition holds exactly in our data:

$$b_{wage}(a) = b_{wage}(c) + b_{rent}(c) \times \delta_{rent\ on\ wage} = 7.093 + 5.910 \times 0.518 = 10.153$$
$$b_{rent}(b) = b_{rent}(c) + b_{wage}(c) \times \delta_{wage\ on\ rent} = 5.910 + 7.093 \times 0.866 = 12.053$$

- The wage coefficient falls by 30% and the rent coefficient by 51%, so part of each univariate slope was the other cost. Collinearity is moderate (VIF = 1.81), so both effects are still precisely estimated.
- **Size check.** If centers simply passed on labor cost, $1,000 more per worker per year is $19.23/week per worker, or about $1.92 per child per week at a 1:10 staff-to-child ratio ($1.28–$2.75 for ratios from 1:15 to 1:7). The estimated $7.09 is several times larger, so the wage variable also captures local affluence and demand. We read the coefficients as associations that predict price, not as causal cost pass-through. That is exactly what the final project needs: a benchmark price given local costs.

Final Project Analysis: Where Should We Locate?

We define an attractive market as one where prices are high *given* what it costs to operate there. The approach has four steps.

- **1. Price benchmark.** For each year 2015–2022 we regress the real (2022 $) median preschool price on the real childcare-worker wage and real median gross rent across metro counties. Estimating each year separately lets the price-cost relationship change over time (for example, during the pandemic).
- **2. Premium.** A county's premium is its actual price minus its predicted price (the residual), in $/week: the price families pay above what local labor and rent costs would predict. It measures pricing power relative to local cost conditions; it is not an estimate of actual operating cost or profit, which the data do not contain. The **score** is the average premium over 2019–2022, so one noisy year cannot carry a county.
- **3. Screens.** (i) a price in 2022 and in at least 3 of 2019–2022; (ii) a positive premium in every observed year 2019–2022; (iii) a county-specific price: the 2022 price is shared by at most 2 counties in the state, and the state-year is not flagged by NDCP as county prices imputed from state-level data (code 3) or propensity-score estimates (code 6). We keep NDCP's estimates for years between state surveys (code 5) and its percentile imputations (code 4); 7 of the 10 picks' 2022 prices are code-5 estimates, so we show the evidence behind each pick below; (iv) at least 2,000 children under 6 whose resident parents are all in the labor force (enough demand to fill a 100+ seat center).
- **4. Pick.** Rank eligible counties by score and keep the best county in each metro area, so that ten picks are ten different labor and rental markets.

Benchmark price model by year (real 2022 $)

Year	N	Intercept	Wage ($000/yr)	Rent ($100/mo)	R^2	RMSE ($/wk)
2015	994	-116.0	8.10	7.86	0.558	37.1
2016	1,019	-109.5	7.83	7.68	0.576	34.9
2017	1,057	-111.3	7.75	8.10	0.599	36.4
2018	1,062	-100.6	7.20	8.24	0.638	33.7
2019	1,061	-92.9	7.03	7.72	0.640	33.5
2020	1,050	-70.4	6.07	7.80	0.622	35.5
2021	905	-105.6	8.17	5.72	0.672	33.6
2022	998	-87.9	7.09	5.91	0.608	37.5

The coefficients are stable across years: $6.07–$8.17 per week for each $1,000 of wage and $5.72–$8.24 for each $100 of rent, with R^2 of 0.56–0.67.

(a) Recommended counties

#	County	Metro area	Price 2022	Pred. price	Prem. 2022	Prem. % of pred.	Score 2019-22	Kids <6, parents work	Conf.
1	Stearns, MN	St. Cloud	$337*	$167	+$170	+102%	+$127	9,982	High
2	Norfolk, MA	Boston	$420	$304	+$116	+38%	+$114	32,889	High
3	McLean, IL	Bloomington	$279*	$168	+$111	+66%	+$95	7,742	High
4	Howard, MD	Baltimore	$336*	$247	+$89	+36%	+$82	16,287	High
5	King, WA	Seattle	$390*	$307	+$83	+27%	+$65	90,718	Medium
6	DuPage, IL	Chicago	$316*	$230	+$86	+37%	+$64	42,574	High
7	Champaign, IL	Champaign	$239*	$174	+$65	+37%	+$49	9,532	High
8	Montgomery, MD	Washington	$336*	$280	+$56	+20%	+$43	54,089	Medium
9	Worcester, MA	Worcester	$285	$243	+$42	+17%	+$43	37,348	Watch
10	Washtenaw, MI	Ann Arbor	$224	$193	+$31	+16%	+$42	13,199	Watch

Prem. % of pred. = 2022 premium ÷ predicted price, using the dollar figures shown. * = 2022 price is an NDCP estimate between state surveys (see the evidence table below). Confidence = number of four alternative scoring rules (log/percent premium, 2022 premium only, survey years only, within-state premium) under which the county stays in the top 10: High = 4, Medium = 3, Watch = 2 or fewer. Funnel: 998 counties scored in 2022 → 973 with enough years → 296 positive every year → 57 with county-specific prices → 44 large enough, in 32 metros.

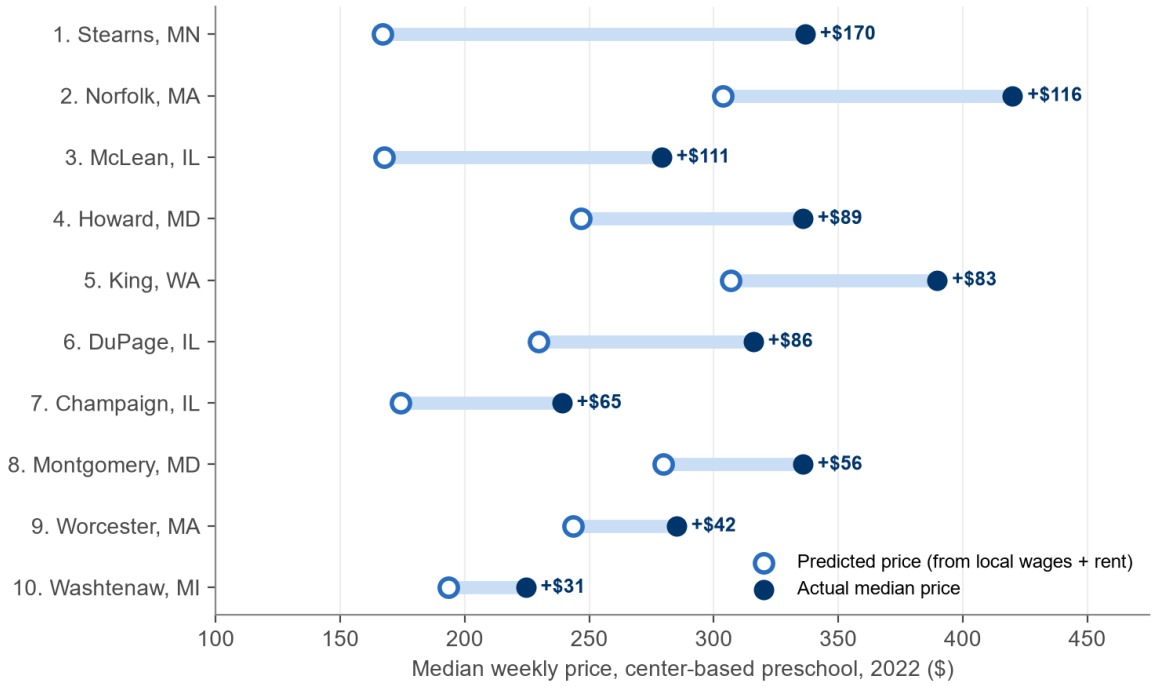
Price evidence behind each pick

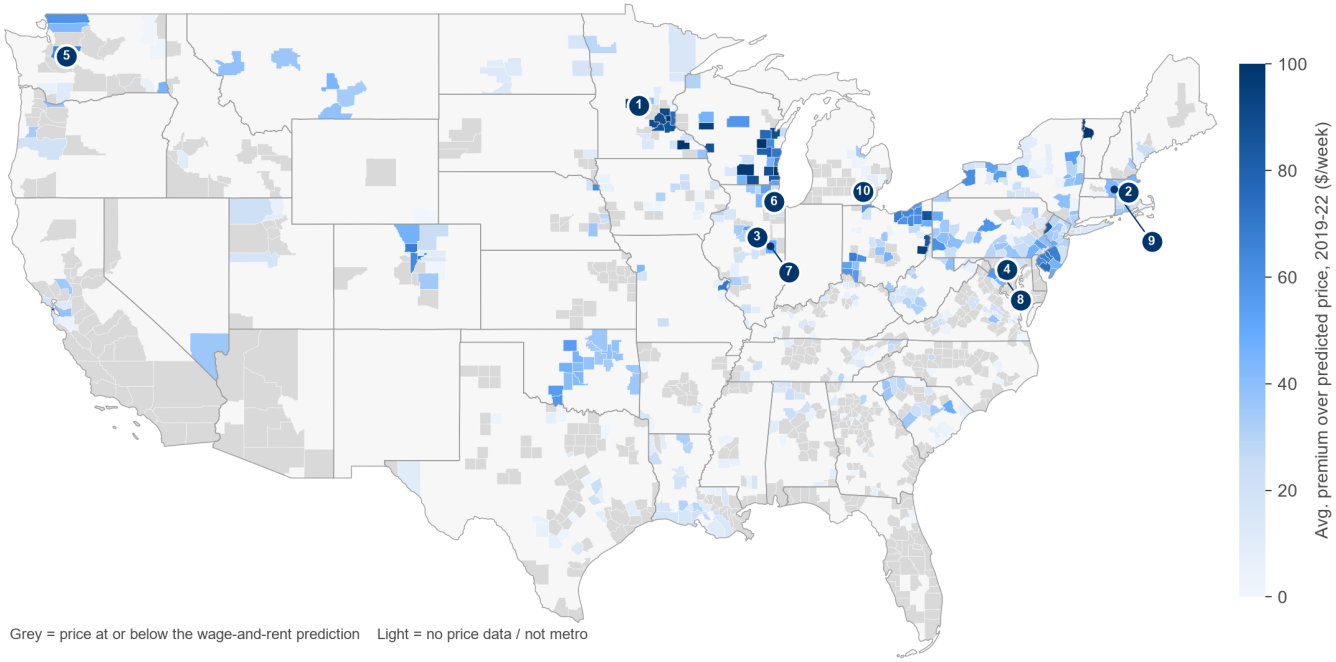
County	NDCP codes, 2022	2022 price is	Latest survey year (no code 5/6): premium	Latest year with no imputation flag (no code 3–6): premium
Stearns, MN	5	an estimate (code 5)	2021: +$135	2021: +$135
Norfolk, MA	2	a survey value	2022: +$116	2022: +$116
McLean, IL	2,4,5	an estimate (code 5)	2021: +$101 (code 4)	2018: +$55
Howard, MD	4,5	an estimate (code 5)	2021: +$79 (code 4)	2017: +$42
King, WA	2,5	an estimate (code 5)	2021: +$65	2021: +$65
DuPage, IL	2,4,5	an estimate (code 5)	2021: +$69 (code 4)	2018: +$21
Champaign, IL	2,4,5	an estimate (code 5)	2021: +$53 (code 4)	2018: +$17
Montgomery, MD	4,5	an estimate (code 5)	2021: +$44 (code 4)	2017: −$7
Worcester, MA	2	a survey value	2022: +$42	2022: +$42
Washtenaw, MI	NA	not documented	2022: +$31 (undocumented)	2022: +$31 (undocumented)

NDCP codes: 2 = converted to a weekly price; 4 = one of the 50th/75th percentiles estimated from the other (NDCP does not say which, so the median may be modeled); 5 = estimate for a year between state surveys; NA = listed as excluded in Appendix B although the workbook has a price (undocumented).

7 of the 10 picks' 2022 prices are NDCP estimates between state surveys, so their 2022 premiums are partly projected. Every pick had a positive premium in its latest survey year. The 5 picks in Illinois and Maryland (McLean, Howard, DuPage, Champaign, Montgomery) rest most on imputed values: their latest surveys carry the percentile-imputation flag, and in their last fully reported surveys (2017–2018) their premiums were −$7 to +$55;

Montgomery (−$7 in 2017) was below its predicted price then. These are the first counties to verify with a field price survey.





Map: average 2019–22 premium for every metro county with price data; numbered markers are the picks.

Beyond the top 10. Next in line (ranks 11–20): Missoula, Montana ($39); Multnomah, Oregon ($39); Anchorage, Alaska ($38); Cascade, Montana ($36); Winnebago, Illinois ($36); Peoria, Illinois ($34); Yellowstone, Montana ($34); Benton, Oregon ($33); Madison, Illinois ($32); Raleigh, West Virginia ($31).

Verify-first list (large, persistent premiums, but the 2022 price is a shared rate-zone price or a state-level or propensity-score imputation, so no county-specific price is observed): Arlington, Virginia ($159, price shared by 1 county, NDCP code 5,6); Olmsted, Minnesota ($121, price shared by 8 counties, NDCP code 5); Winnebago, Wisconsin ($119, price shared by 10 counties, NDCP code NI); Milwaukee, Wisconsin ($115, price shared by 10

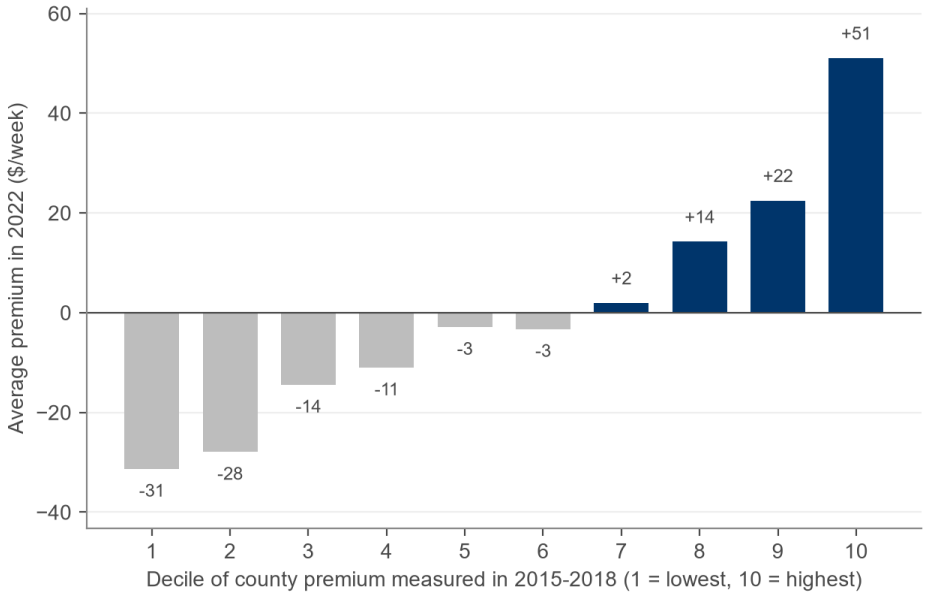
counties, NDCP code NI); Ramsey, Minnesota ($100, price shared by 8 counties, NDCP code 5); Dane, Wisconsin ($97, price shared by 10 counties, NDCP code NI); Belmont, Ohio ($96, price shared by 29 counties, NDCP code NI); Eau Claire, Wisconsin ($95, price shared by 10 counties, NDCP code NI); Brown, Wisconsin ($89, price shared by 10 counties, NDCP code NI); San Francisco, California ($89, price shared by 1 county, NDCP code 5,6).

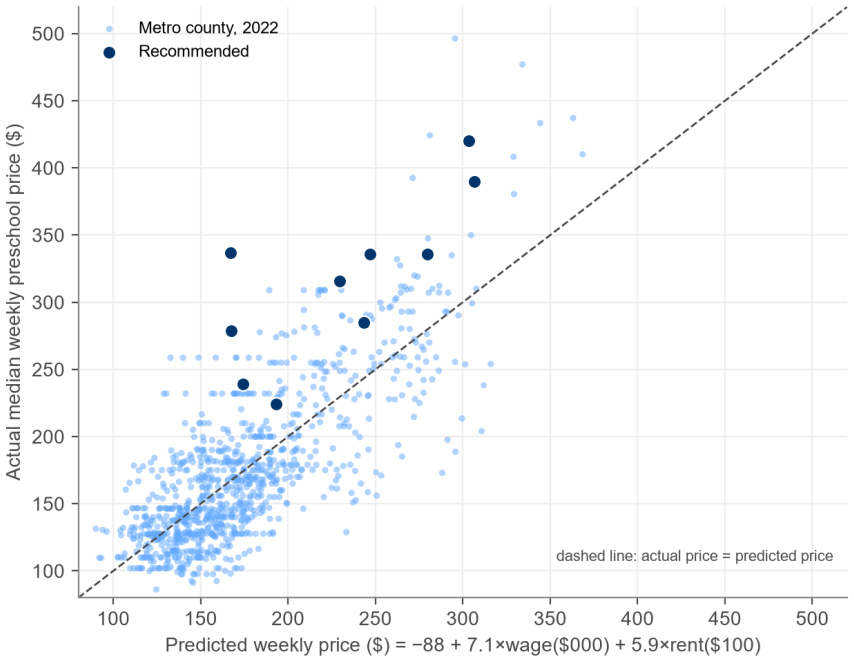
(b) Reliability of the prediction

Model (2022)	In-sample R^2	Out-of-sample R^2	Out-of-sample RMSE ($/wk)
Mean only	0.000	-0.006	59.9
Wage only	0.547	0.541	40.4
Rent only	0.460	0.447	44.4
Wage + rent (benchmark model)	0.608	0.598	37.9
Log wage + log rent	0.580	0.602	37.7
Benchmark + demand characteristics	0.683	0.671	34.3

Out-of-sample fit is grouped 5-fold cross-validation: whole OEWS metro areas are held out, so counties that share a wage never appear in both training and test folds.

- **Little sign of overfitting.** The benchmark model loses almost nothing out of sample (R^2 0.608 → 0.598). A typical prediction error is about $38/week. 8 of the 10 picks have 2022 premiums at least 1.5 times that ($56–$170). Worcester ($42) and Washtenaw ($31) are within about one typical error, so their premiums are the hardest to tell from noise; they are also the two picks rated Watch.
- **Back-test.** We reran the whole rule as of 2018 (scores from 2015–2018, same screens) and checked 2022. Counties in the top decile of the 2015–18 premium averaged $51/week above their predicted price in 2022; the bottom decile averaged -$31 (correlation 0.64; slope 0.69). Of the 10 counties picked in 2018, the 6 that still had 2022 price data averaged $70/week above their predicted price, and all 6 were above it. (The other 4 are in Pennsylvania and Connecticut, which have no usable 2022 county prices.)





Actual vs predicted 2022 price. Counties above the dashed 45° line charge more than their local wages and rents predict; the picks (dark) are well above it.

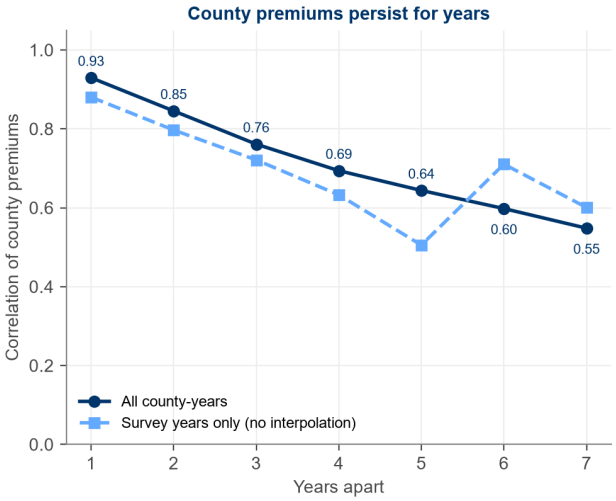
(c) Market characteristics that predict promising locations

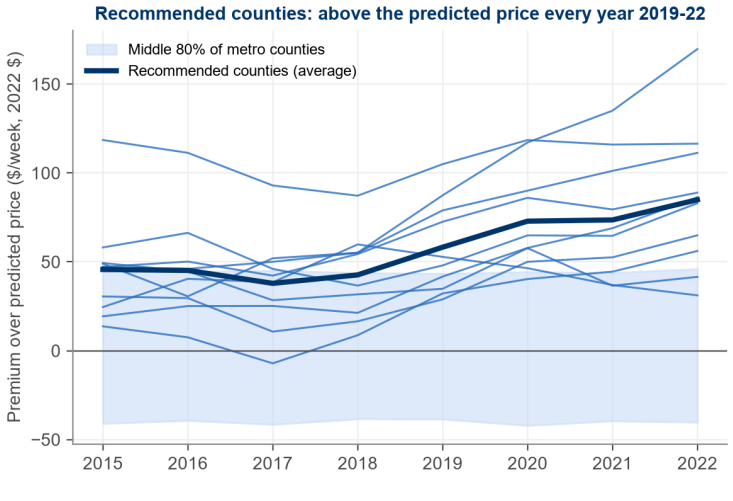
Characteristic	Coef ($/wk)	SE (clustered)	p	Standardized ($/wk per SD)	Within-state std. (p)
Per-capita income ($10k)	4.93	3.13	0.1151	4.67	2.88 (0.5225)
Women 25+ with bachelor's+ (per 10 pp)	5.19	2.20	0.0184	5.74	6.64 (0.0092)
Women 16–64 full-time year-round (per 10 pp)	-0.80	2.41	0.7393	-0.45	-1.53 (0.2039)
Under-6 needing care, % of pop (per 1 pp)	3.32	1.52	0.0293	2.97	-0.56 (0.6806)
log population	-0.54	1.14	0.6349	-0.78	1.04 (0.5495)

Regression of the 2022 premium on demand-side ACS characteristics, N = 998, R^2 = 0.072. The last column adds state fixed effects.

- **Female college share** is the strongest and most robust predictor of a premium: about $5.74/week per standard deviation, significant, and larger within states ($6.64). The share of young children who need care is positive (p = 0.029), but not within states. Income is positive but imprecise once education is held fixed.
- Together these explain only 7% of the premium. Demand traits point to where premiums tend to be, but they do not replace measuring the premium directly, so the measured premium is our primary screen.
- **Profile of the picks (2022 means vs all metro counties):** price $316 vs $174; per-capita income $52,332 vs $37,859; women with a bachelor's+ 51% vs 31%; under-6 children with working parents 31,436 vs 11,538.
- **Characteristics we use:** (1) price premium over the wage-and-rent benchmark; (2) persistence of that premium; (3) county-specific prices; (4) market depth (≥ 2,000 under-6 children with working parents); and, to prioritize the next wave of markets, (5) a high share of college-educated women and higher income.

(d) Are premiums stable over time?





Years apart	1	2	3	4	5	6	7
Correlation, all county-years	0.93	0.85	0.76	0.69	0.64	0.60	0.55
Correlation, survey years only	0.88	0.80	0.72	0.63	0.51	0.71	0.60
Pairs (all / survey)	6,787 / 1,045	5,883 / 1,463	4,828 / 1,008	3,769 / 762	2,710 / 887	1,673 / 407	903 / 260

- Premiums are highly persistent: 0.93 one year apart, still 0.55 seven years apart. NDCP interpolates some years between state surveys, which could create persistence mechanically, so we repeated the calculation with survey years only. The pattern is similar.
- Of counties in the top fifth of premium in 2015–18, 67% were still in the top fifth in 2019–22 and 88% in the top two-fifths (correlation of the period averages 0.76). 80% of counties had the same sign of premium in 2019 and 2022.
- Our picks were above their predicted price in every year 2019–2022 (a screen). Over 2015–2018 they averaged about $43, and one dipped slightly below zero in 2017 (-$7). Their premium has widened since 2019.

(e) What our predictions are missing

- **Competition and capacity:** licensed-center counts, waitlists and vacancy rates. A premium can signal a shortage that competitors will fill.
- **Public pre-K and subsidies:** state pre-K expansion and CCDF voucher reimbursement rates can cap prices, especially for 4-year-olds.
- **State regulation and full labor cost:** staff-to-child ratios, group sizes, licensing and benefits. 43% of the variance in 2022 premiums is between states. Some of that is regulation and some is differences in state survey methods; the within-state check keeps 9 of our 10 picks.
- **Commercial real estate:** residential median gross rent is only a proxy for lease and build-out costs.
- **Price data quality:** NDCP prices come from state market-rate surveys, some interpolated or published by rate zone. 7 of our 10 picks' 2022 prices are estimates between state surveys (e.g. Stearns County's is extrapolated from Minnesota's 2021 survey). A field price check is the first next step.
- **Demand trajectory:** births, under-5 population growth, migration, remote work and income growth (we use levels, not trends), plus the 75th-percentile price as a gauge of the premium segment.